

Intro Machine Learning

EDSD workshop on Machine Learning

Andrea Nigri, PhD

June 11, 2025

University of Foggia,
andrea.nigri@unifg.it

BACKGROUND

Classical Inference: The Parametric Approach

- Classical statistical inference is based on the assumption that data are generated according to a probabilistic model.
- We choose a **family of parametric models**, typically defined by a small set of parameters.
- The model is a mathematical abstraction—an idealization of the real process that generates our data.

Parametric Models: A Probabilistic Assumption

- For example, we might assume that our data are drawn from a Normal distribution with unknown mean and variance.
- **This is a strong assumption:** we are claiming that all the randomness in our data can be explained by this model, up to some unknown parameters.
- Our goal is to use the data to estimate these parameters and to make inferences about the underlying process.

The Classical Inference Pipeline

1. **Specify** a family of models based on probabilistic assumptions (e.g., Normal, Binomial, Poisson).
2. **Estimate** the parameters of the model from the observed data (e.g., using Maximum Likelihood).
3. **Evaluate** the fit of the model and test hypotheses about the parameters.

Key Point

In classical inference, all uncertainty and variability are formalized through the lens of a parametric probabilistic model.







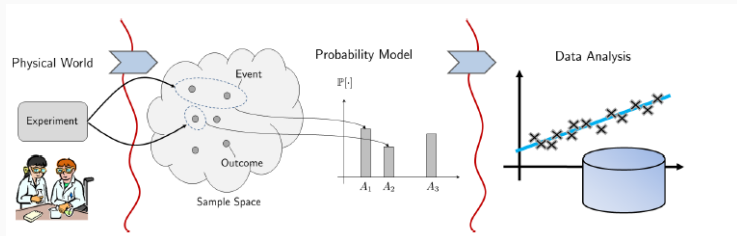
The bricks, i.e. the individual elements build a model, representative of a reality.

- In the **data analysis**, these bricks are the data
- Data that must be treated according to a given pattern - our statistical model - to recreate/represent an observed reality that we want to approximate.

Why....? to understand an explicit or hidden pattern inherent in the data ... **why try to understand a pattern?**

Interpret or predict

- To make decisions under conditions of uncertainty or even to anticipate upcoming scenarios and events.
- To this end, statistics uses probability to develop models with an underlying probabilistic nature to explain a given reality.



Strengths and Limitations

- **Strengths:** The parametric approach is mathematically elegant and often computationally efficient. It allows precise statements about uncertainty.
- **Limitations:** If the true data-generating process is far from our assumed model, our inferences may be misleading.

All models are wrong,
but some are useful.

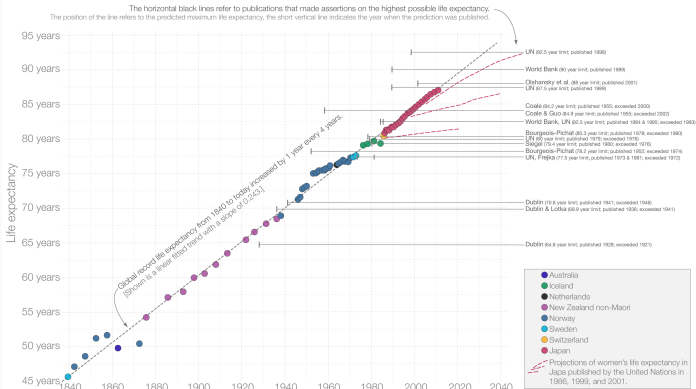
George Box, British statistician (1919 – 2013)

We will
never the true data-generating model, i.e., the real process by
nature that generated the data.”

Linear model

Record life expectancy of women from 1840 to the present

Shown is the highest known life expectancy of women at each point in time and the country that achieved that level of population health.



This chart was originally published in Oppen and Vaupel (2002) – Broken Limits to Life Expectancy, Published in Science, 296, 5570, 1029-1031.

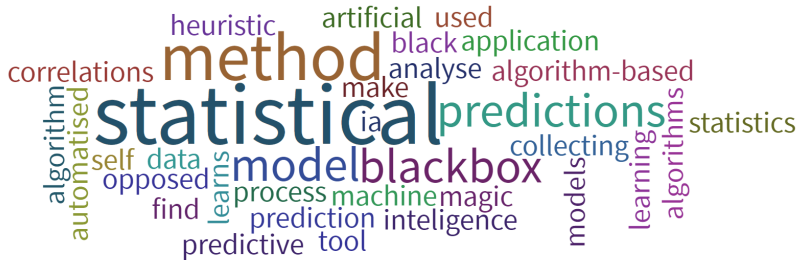
This version of the chart is extending Oppen and Vaupel (2002) by adding more recent estimates for Japan and is completely redrawn and newly annotated. Published under CC-BY-SA by www.OurWorldinData.org

What's Machine (Deep) learning

What is Machine Learning?

- Just one or...more definitions?
- **Let's check class opinion!**

Machine Learning Definition:



What is deductive learning?

Aristotele (384 aC, 322 aC)

General rule to **Specific case**

- General rule: *All man are mortal*
- *Socrates is a man*
- Specific case: *Socrates is mortal*

What is inductive learning?

Leonardo da Vinci; Galileo Galilei; ...; Isaac Newton.

Specific case to **General rule**

- Specific case: *Socrates was a man*
- *Socrates died*
- General rule: *all men are mortal*

Inductive learning-Machine learning

- **generalization**, with which we get information about a group of things, people,...by examining a portion - or **sample** - of that group.
- BIG DATA (5 V's), the concept of the sample has been overcome, obtaining the entire population.

...Starting from here, we can get a fairly accepted view of Machine Learning

Machine learning

- It is a **system** that, starting from some examples during the training phase, is able to **learn**. Then it is able to **generalize** and manage new data (in the same application area).
- learning from mistakes: One of the best applications is by Turing who deciphered the Enigma code during the world war

APPLICATION AREAS:

- Identify the risk factors for a specific cancer.
- Predict whether someone will have a heart attack on the basis of demographic, diet and clinical measurements.
- Customer segmentation
- Customers churn rate

Types of learning algorithms

Unsupervised learning:

- We do not have a target variable.
- Examples: PCA, Cluster Analysis, Autoencoder Neural Network

Supervised learning:

- We have a target variable Y , which we want to predict in relation to explanatory variables X .
- Examples: Linear Regression, Logistic Regression, Regression tree, Neural Network

Supervised Learning

According to the statistical learning theory (SLT).

- Given a set of training data $D = \{(\mathbf{x}_1, y_1) \dots (\mathbf{x}_1, y_1)\}$ in $\mathbb{R}^n \times \mathbb{R}$ sampled according to unknown probability distribution $P(\mathbf{x}, y)$,
- Consider a Loss function $L(y, \mathbf{f}(\mathbf{x}))$ that measures the error done by $\mathbf{f}(\mathbf{x})$, for a given $\mathbf{x}$,

The problem consists in finding a function f that minimizes the expected error:

$$\int L(y, \mathbf{f}(\mathbf{x}))P(\mathbf{x}, y)d\mathbf{x}dy$$

Supervised Learning

- "distribution-free" approach
- The problem consists in finding a function f that minimizes the expected error:

$$\int L(y, f(\mathbf{x}))P(\mathbf{x}, y)d\mathbf{x}dy$$

- $P(\mathbf{x}, y)$ is unknown, we need to use some induction approach.
- We used is Empirical Risk Minimization (ERM) over a set of possible functions, called hypothesis space. Formally this can be written as minimizing the empirical error:

$$\frac{1}{l} \sum_{i=1}^l L(y_i, f(\mathbf{x}_i))$$

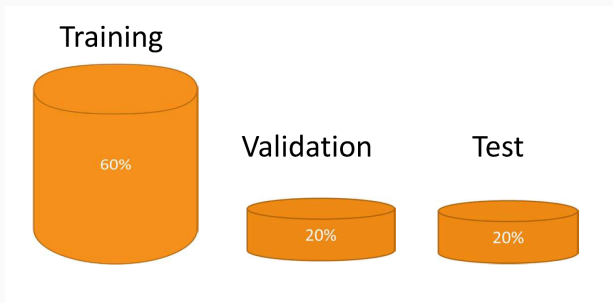
My First ML Project

- Load the dataset and try models.
- Model evaluation on the same dataset?
- More data to train the model is better, right?

...Wrong!

Model training

- The drawback: it is not able to generalize to new data.
- A split of data 70-30 for training to test datasets is a good start.



The search for the optimal parameters is then carried out through an optimization process. The iterations of the algorithm lead to the optimization of the parameter and the minimization of the error.

Statistics vs. Statistical Learning up to Machine Learning

Statistics vs. Statistical Learning up to Machine Learning

Statistical Learning: New term...old concepts.

- At the beginning of the nineteenth century - least squares
- 1940s - logistic regression
- early 1970s, generalized linear model

... they were almost linear methods because fitting non-linear relationships was computationally difficult.

Non-linear methods

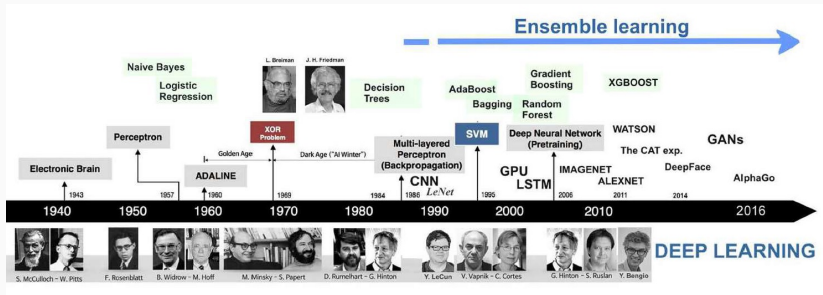
...By the 1980s, computing technology had finally improved sufficiently, that non-linear methods were no longer computationally prohibitive.

- mid 1980s, regression trees and generalized additive models ...
Neural networks gained popularity
- 1990 support vector machines
- Machine Learning: modern evolution of statistical learning

Machine Learning vs. Deep Learning

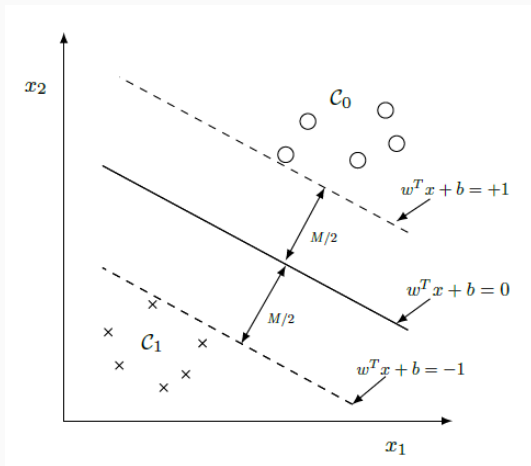
- While all deep learning is machine learning, not all machine learning is deep learning
- Machine Learning help the computer learn how to recognize things. This training requires a significant amount of human effort.
- Deep learning algorithms: hierarchical models (multi-layered neural network) that do not require preprocessing.

Time line



Support Vector Machine

- SVM by Vapnik et al., (1997); popular among classification problems.
- Offers a optimal solution identifying a linear hyperplane which correctly classifies observed data and lies as far as possible from them.



- Problem of binary classification
- I observation of two explanatory variable X_1, X_2 and the realisation of a target class Y , are given as:

$$(x_1, y_1), (x_2, y_2), \dots, (x_I, y_I), x \in \mathbb{R}^n, y \in \{+1, -1\}$$

- find the optimal separating function without knowing the underlying probability distribution $P(x, y)$.

By using training examples, our machine finds parameters $\mathbf{w} = [w_1 w_2 \dots w_n]^T$ and b of decision function $d(x, w, b)$ given as

$$d(x, w, b) = \mathbf{w}^T \mathbf{x} + b = \sum_{i=1}^n w_i x_i + b$$

where $\mathbf{x}, \mathbf{w} \in \mathbb{R}^n$, and the scalar b is called a bias. After the successful training stage, the learning machine, given unseen pattern x_p , produces output according to the following decision rule:

- if $d(x_p, w, b) > 0$, the pattern x_p belongs to $\{C_0\}$ (i.e. , $y_1 = +1$) and
- if, $d(x_p, w, b) < 0$ the pattern x_p belongs to $\{C_1\}$ (i.e., $y_2 = -1$).

Where $\{C_0, C_1\}$ are the two possible output classes and b is a bias term

Where the two dashed lines describe in Figure 28, the so-called *margin* to be maximized, are equal to

$$M = \min_{\mathbf{x}: y=1} \frac{\mathbf{w}\mathbf{x}}{\|\mathbf{w}\|} - \max_{\mathbf{x}: y=-1} \frac{\mathbf{w}\mathbf{x}}{\|\mathbf{w}\|}, \quad (1)$$

Since from the decision rule and (1) follows that

$$M = 2 \frac{|\mathbf{w}\mathbf{x}|}{\|\mathbf{w}\|} = \frac{2}{\|\mathbf{w}\|} \quad (2)$$

For ease of calculation we can write the SVM optimisation problem as

$$\begin{aligned} \min \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{sub} \quad & y_n [\mathbf{w}^T \mathbf{x}_n + b] \geq 1 \quad \forall n = 1, \dots, N \end{aligned} \quad (3)$$

Regression Tree

Model:Regression Tree

Decision trees can be applied to both regression and classification problems.

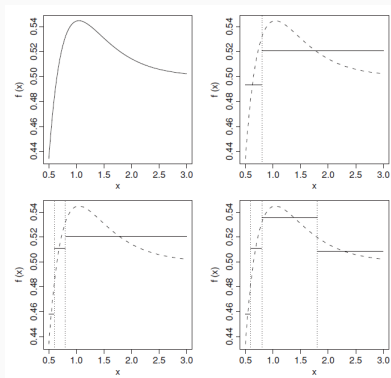
We can use Regression Tree for getting the best approximation of a function.

$$y = f(x) \tag{4}$$

with $x \in \mathbb{R}$ is to use a step function, that is, a piecewise constant function. **But...**

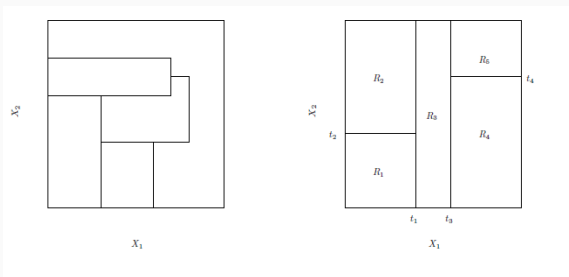
Model:Regression Tree

- (a) how many x-axis subdivisions?
- (b) where are the subdivision points?
- (c) which value of y to consider in each interval?



Model: Regression Tree

- Predictor space division
- $X_1, X_2, \dots, X_p$, into J distinct and non-overlapping regions, $R_1, R_2, \dots, R_J$.



Model:Regression Tree

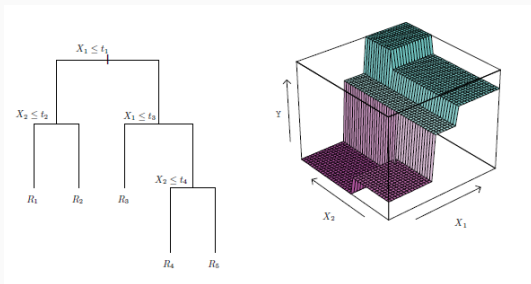
- The regions shape: high-dimensional rectangles, or boxes.
- The goal is to find boxes $R_1, \dots, R_J$ that minimize the RSS, given by

$$\sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2 \quad (5)$$

- Once the regions $R_1, \dots, R_J$ have been created, we predict the response for a given test observation using the mean of the training observations in the region to which that test observation belongs.

... Computational drawbacks

- **Top-Down** approach, using a recursive binary splitting.
- It begins at the top of the tree.
- then it splits the predictor space; each split is indicated via two new branches.

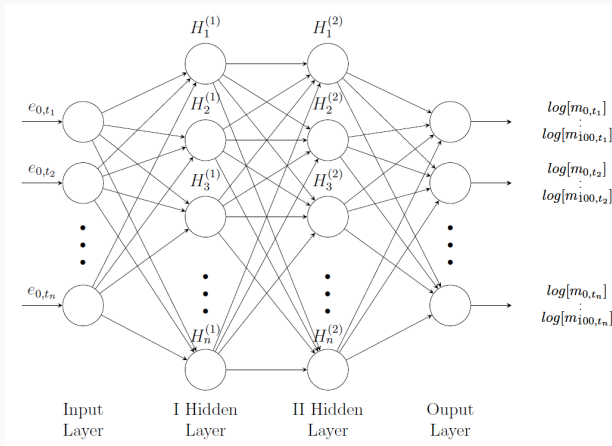


- if $X_1 < t$ then the unit belongs to the left branch. (the cutpoint t such that the space predictor splitting leads to the greatest possible reduction in RSS.)
- otherwise it belongs to the right branch
- To identify the best split: ENTROPY or INDEX OF GINI
- The maximum purity is when only one class of Y is present in the node

Model: Random Forest

- Regression Trees: low bias and high variance estimations
Indeed, Random forests basically consist in building an ensemble of decision trees grown from a randomized variant of the tree.
- Introduce a random perturbation among the trees and combine the predictions of all these trees using aggregation techniques.
- Breiman 1996: bagging by using random bootstrap copies of the original data.

Deep Learning: Deep Neural network



Deep Learning: Neural Network

NN training involves an unconstrained optimization problem where the aim is to minimize a function in high dimensional space the so-called loss function, that measures the difference between the predicted values and observed ones. The back-propagation is the most used algorithm for the training of NNs. The algorithm compares the predicted values against the desired ones (objective) and modifies the synaptic weights by back-propagating the gradient of the loss function. Schematically, the procedure alternates forward and backward propagation steps:

- in the forward step, the prediction is computed fixing the synaptic weights,
- in the backward step, the weights are adjusted in order to reduce the error of the network.

The NN iteratively performs forward and backward propagation and modifies the weights to find the combination that minimizes the loss function

$$\mathcal{L}(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

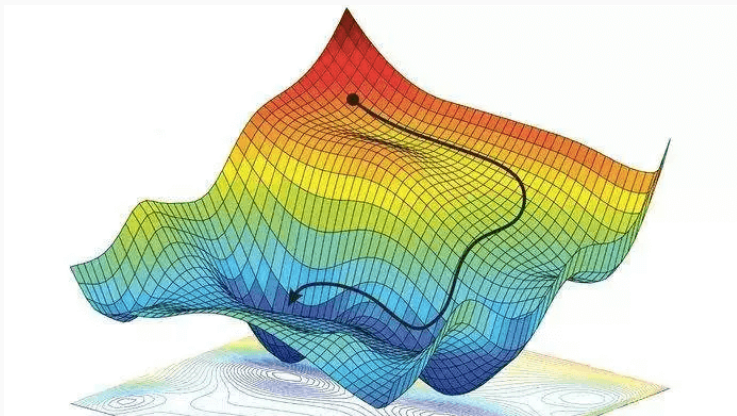
To minimize the loss function, we use the Gradient Descent optimization algorithm, which proceeds by minimizing $\mathcal{L}$ by differentiating the loss function with respect to the weights ($\mathbf{W}$).

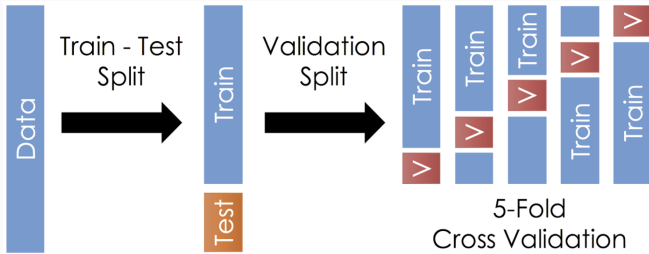
The algorithm proceeds using the chain derivation rule described in the following equation:

$$\frac{\partial \mathcal{L}(y, \hat{y})}{\partial w_{n,n}^{(k)}} = \frac{\partial \mathcal{L}(y, \hat{y})}{\partial H_n^{(k)}} \frac{\partial H_n^{(k)}}{\partial z_n^{(k)}} \frac{\partial z_n^{(k)}}{\partial w_{n,n}^{(k)}} \quad (6)$$

where $z_n^{(k)} = w_n^{(k)} H_n^{(k-1)} + b_n^{(k)}$. To update the weights ($\tilde{\mathbf{W}}$), the gradient of the loss function, $\nabla \mathcal{L}_t(y, \hat{y})$, is multiplied by a scalar, η , often called learning rate, according to the following scheme:

$$\tilde{\mathbf{W}} = \mathbf{W} - \eta \nabla \mathcal{L}_t(y, \hat{y}) \quad (7)$$





- The search for the optimal parameters is then carried out through an optimization process where the NN initial weights are selected in an arbitrary (random) way so they are not optimal parameters. The iterations of the algorithm lead to the optimization of the weights and minimization of the error. The choices concerning the type of architecture (e.g., the number of hidden layers, units for each layer) and the hyperparameter (e.g., learning rate, activation functions, and loss function), remains a heuristic problem for NN users: the choice often depends on the type of data and it is not an easy step.

